

Fundamentals and recent advances in tuberculosis treatment and prevention

Lilian Tina Minja
MD, MMed Internal Medicine, PhD Epidemiology

Global burden of tuberculosis (TB)

- TB is the leading cause of death from a single infectious agent globally
- In 2024: 10.7 million people developed TB; 1.2 million deaths (150,000 among PLHIV)
- Incidence: 131/100,000 globally

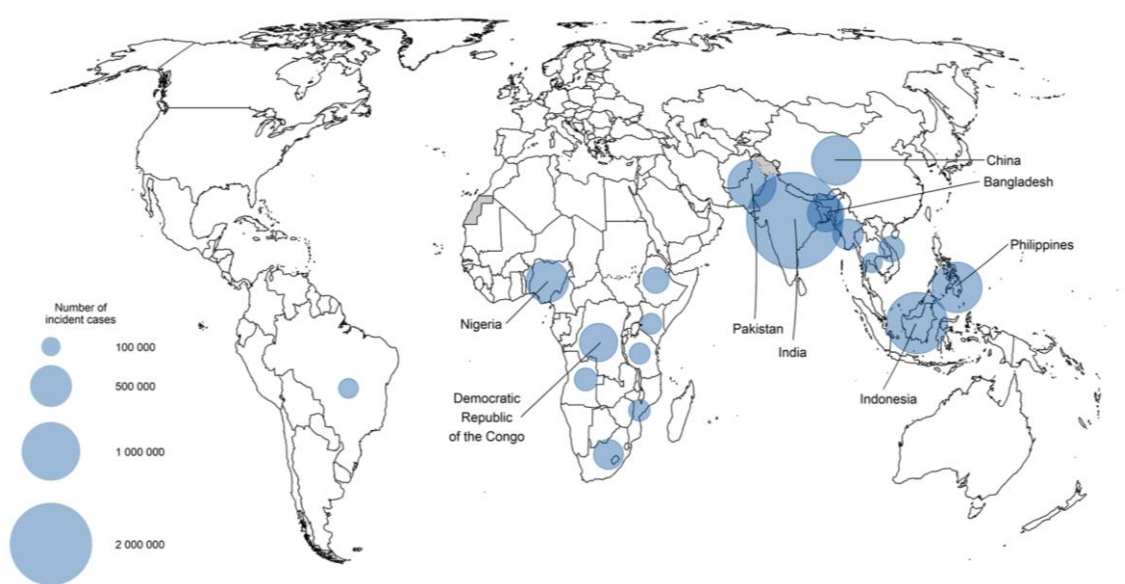
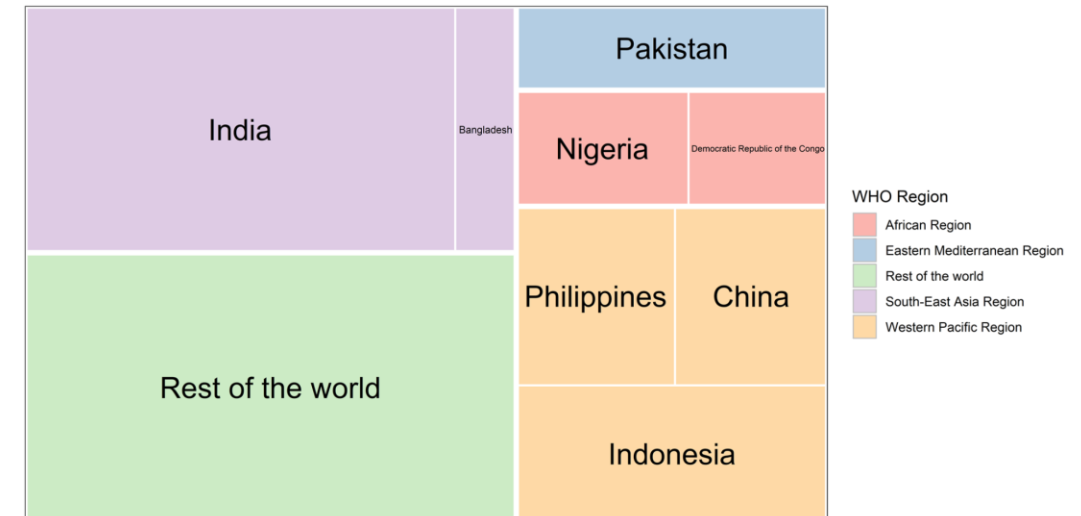


Fig: Estimated number of incident TB cases for countries with at least 100 000 incident cases, 2024



WHO Global TB report, 2025

Burden of DR-TB, 2024

- Developed RR-TB: 390,000
- Treated: 164 545 (42%)

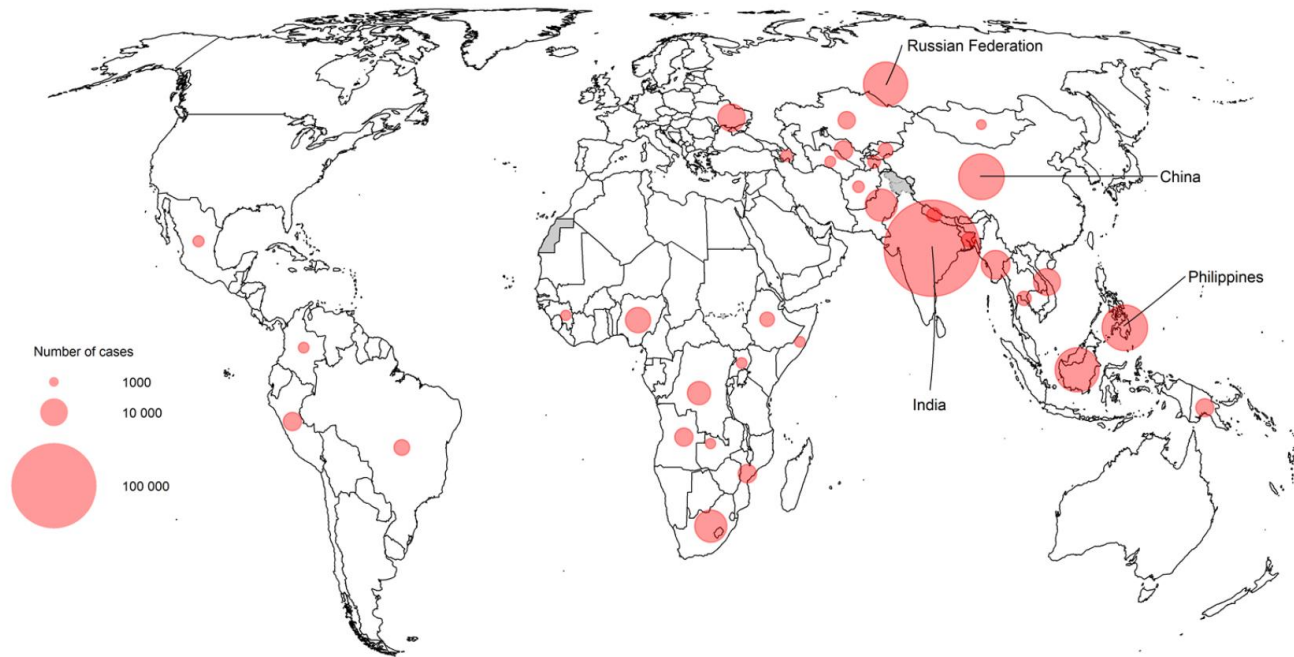
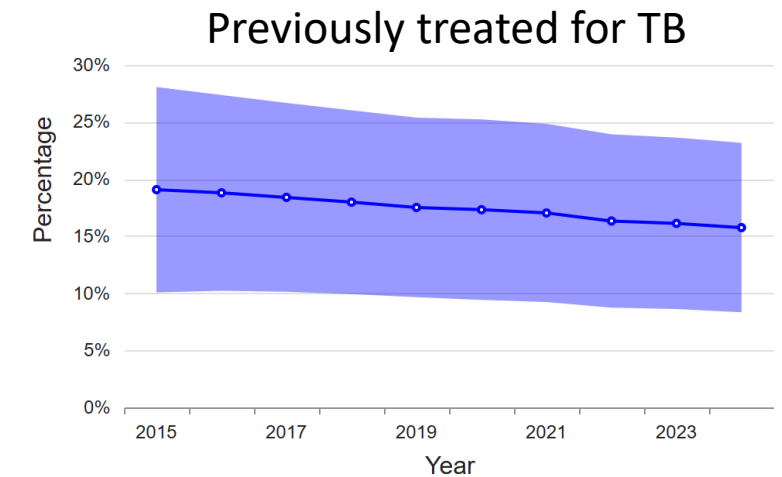
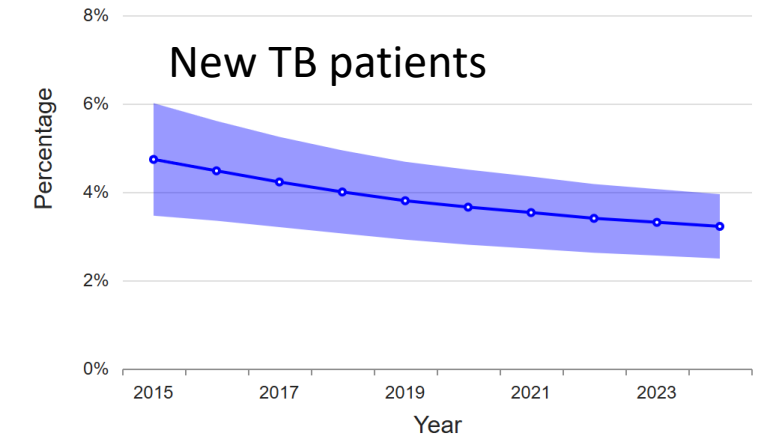
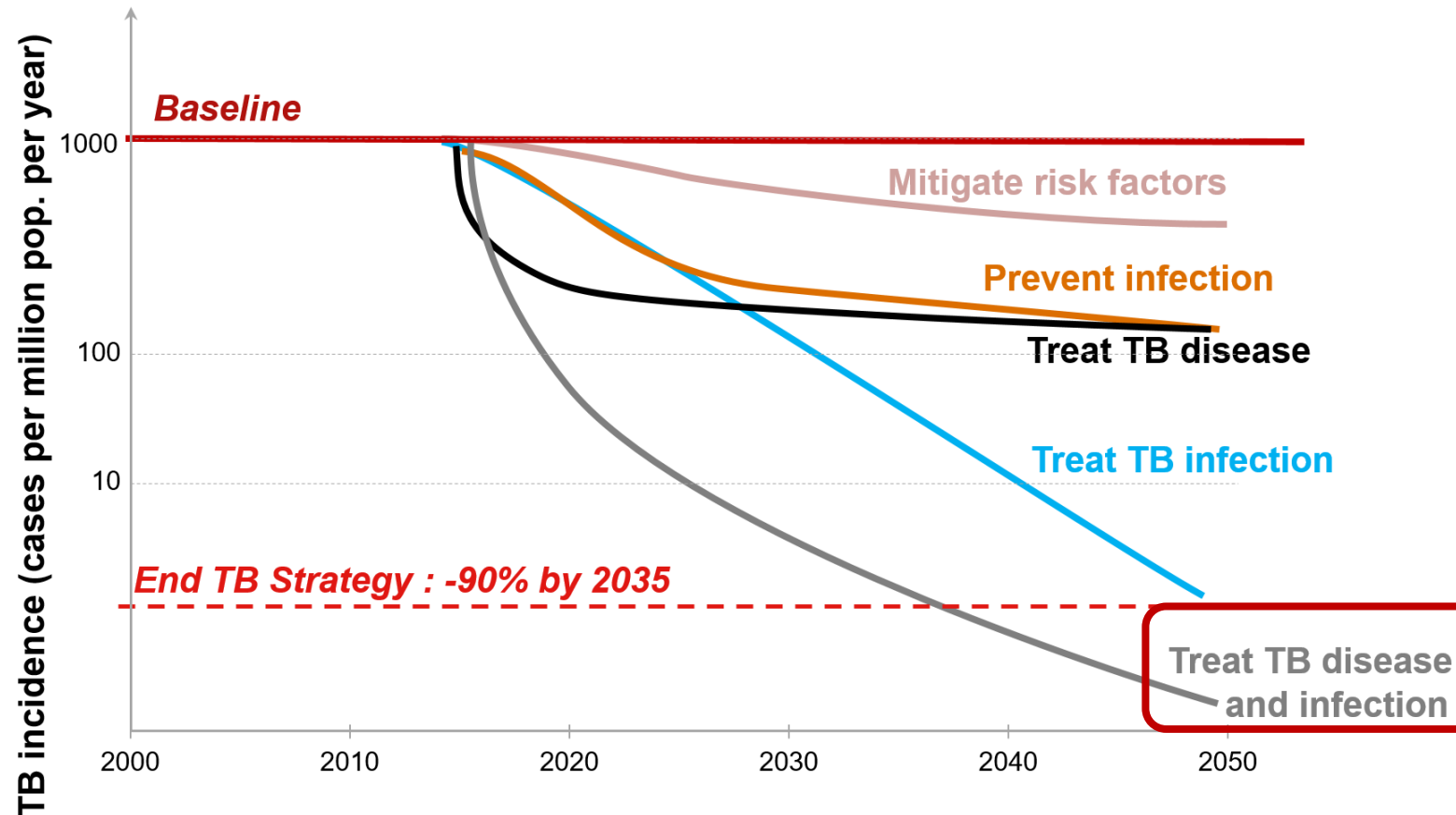


Fig: Burden of MDR/RR TB (incident cases) - countries with at least 1000 incident cases in 2024
WHO Global TB report 2025



WHO Global TB report, 2025

Expected impact of different interventions



From Dye C et al., *Prospects for Tuberculosis Elimination*. *Ann Rev Public Health* 2013. 34:271-86

TB treatment

Why do we need so many drugs?

Why does treatment take so long?

Combination therapy and treatment duration in TB: scientific rationale



Spontaneous mutation
→ drug resistance



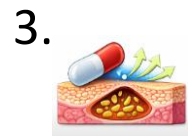
High bacillary load and
cavitary disease



Heterogeneous bacillary
populations (different
metabolic states)



Slow growth rate of
Mycobacterium tuberculosis



Pharmacokinetics and
drug penetration into lesions



Host immunity plays a
critical role

Fundamental principles of TB treatment

Combination drug therapy

- Prevents development of drug resistance
- Ensures more effective cure of the patient

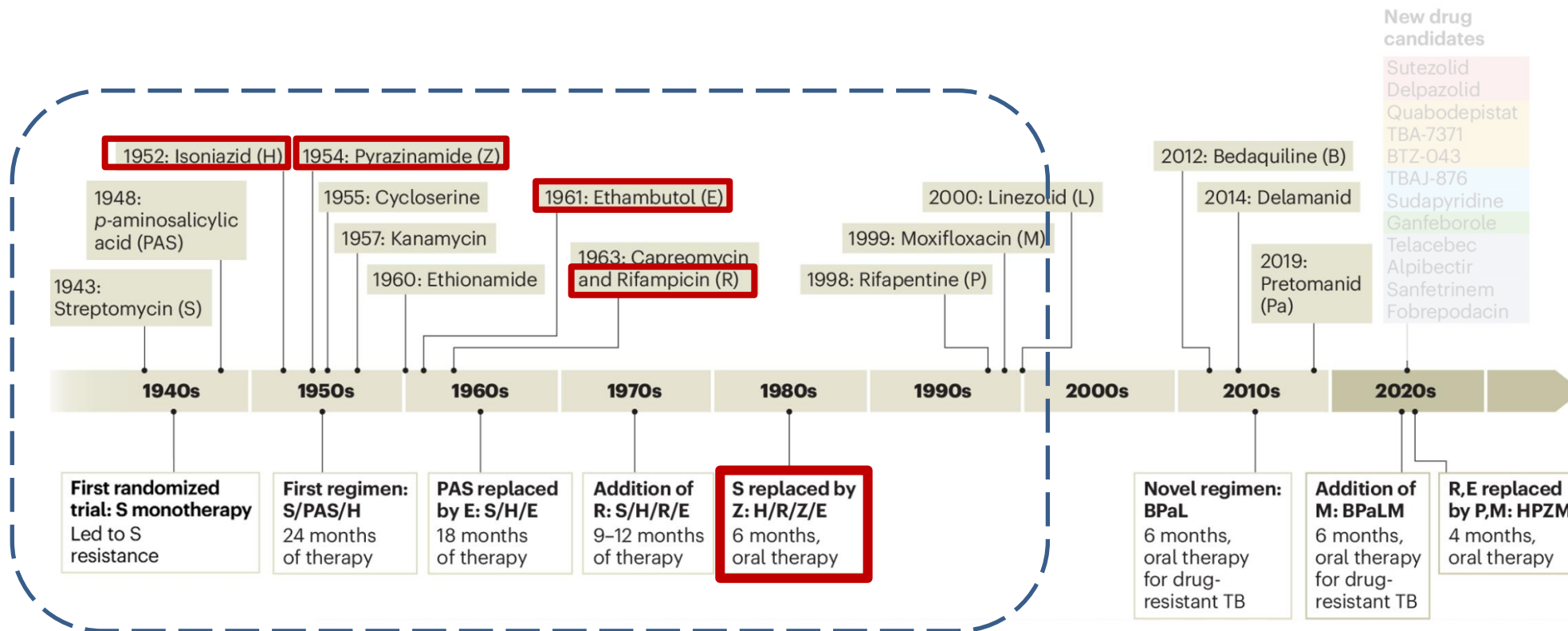
Optimal duration of therapy with good adherence

- Prevents relapse and treatment failure
- Supports cure of the individual
- No symptoms $\neq$ treatment complete

Monitor for adverse drug reactions and drug interactions

- Ensures patient safety
- Prevents complications $\rightarrow$ treatment failure
- Especially critical in PLHIV (ART interaction)

Evolution in TB treatment drugs



- First major breakthrough 1940s
- Currently used regimen for DS TB (1980s) 2RHZE/4RH
- More than 60 years later, and despite challenges in its use, this remains to be the first line treatment of DS TB.

Nuermberger, E.L., Chaisson, R.E. Restocking the tuberculosis drug arsenal. *Nat Med*, 2024

Treatment of DS TB

- New patients with pulmonary TB should receive a regimen containing 6 months of rifampicin **2HRZE/4HR**
- People ≥ 12 years with DS PTB, may receive a 4-month regimen **2HPMZ/2HPM**
- Children and adolescents between 3 months and 16 years of age with non-severe TB (without suspicion or evidence of MDR/RR-TB) should use **2HRZ(E)/2HR**

H - isoniazid, **R** - Rifampicin, **Z** - Pyrazinamide, **E** - Ethambutol, **P** - Rifapentine, **M** - Moxifloxacin

Treatment of DS TB using 4 months regimen

- People ≥ 12 years with DS PTB, may receive a 4-month regimen

2HPMZ/2HPM

The NEW ENGLAND JOURNAL of MEDICINE

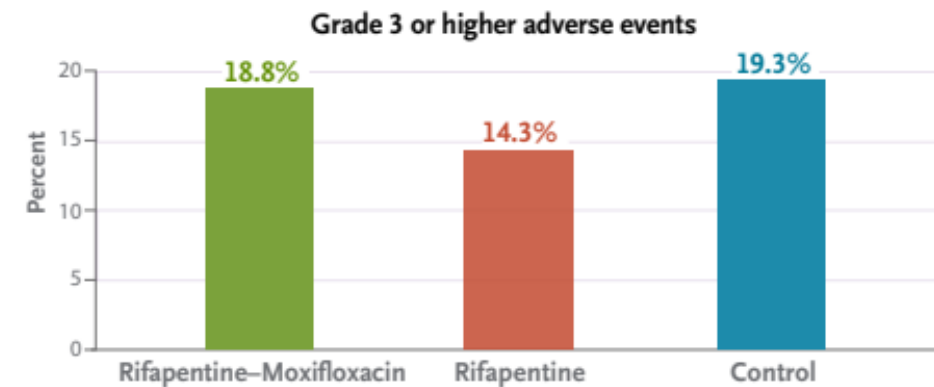
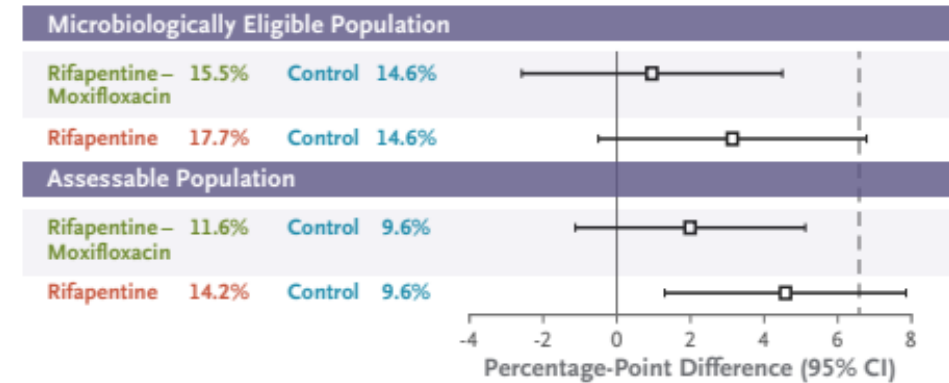
ORIGINAL ARTICLE

Four-Month Rifapentine Regimens with or without Moxifloxacin for Tuberculosis

S.E. Dorman, P. Nahid, E.V. Kurbatova, P.P.J. Phillips, K. Bryant, K.E. Dooley,

N Engl J Med 2021;384:1705-18.

DOI: 10.1056/NEJMoa2033400



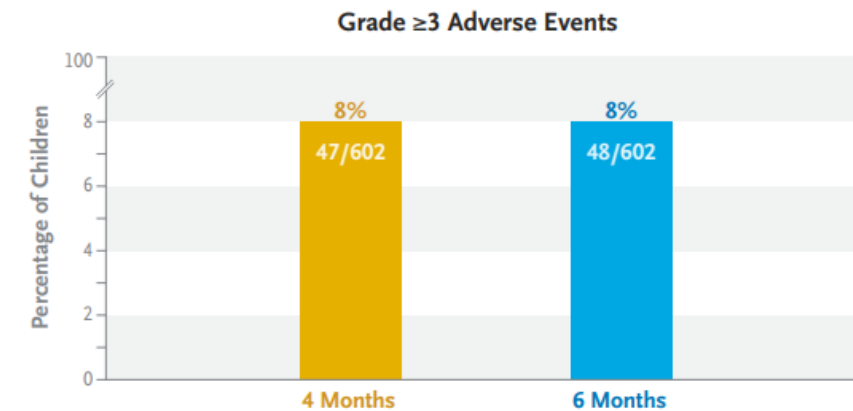
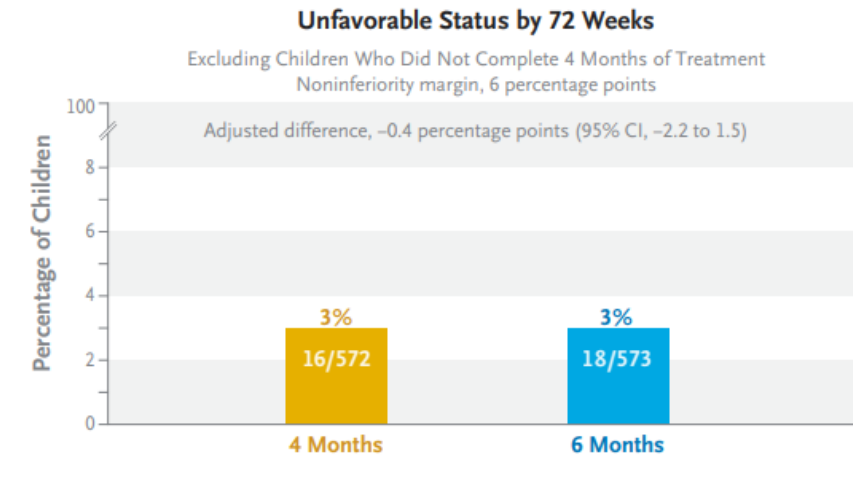
Treatment of DS TB using 4 months regimen

- In children and adolescents between 3 months and 16 years of age with non-severe TB (without suspicion or evidence of MDR/RR-TB): **2HRZ(E)/2HR**



Shorter Treatment for Nonsevere Tuberculosis in African and Indian Children

A. Turkova, G.H. Wills, E. Wobudeya, C. Chabala, M. Palmer, A. Kinikar, S. Hissar, L. Choo, P. Musoke, V. Mulenga, V. Mave, B. Joseph, K. LeBeau, M.J. Thomason, R.B. Mboizi, M. Kapasa, M.M. van der Zalm, P. Raichur, P.K. Bhavani, H. McIlleron, A.-M. Demers, R. Aarnoutse, J. Love-Koh, J.A. Seddon, S.B. Welch, S.M. Graham, A.C. Hesselting, D.M. Gibb, and A.M. Crook, for the SHINE Trial Team*



DS-TB treatment regimens



Fig: Adapted from overview of WHO consolidated guidelines on TB treatment, 2025

Treatment options for DS-TB patients

2HRZE/4HR

- Considered safe for pregnant woman;
- Can be used in all age groups;
- Can be used regardless of HIV status or disease severity.

2HPMZ/2HPM

- Applicable for ages 12 years old and above
- Without severe extrapulmonary TB
- Can be used with PLHIV with a CD4 count > 100 cells/mm³;
- Pregnant, breastfeeding and postpartum women cannot benefit from this regimen

2HRZ(E)/2HR

- For children and adolescents between 3 months and 16 years of age
- with non-severe TB
- without suspicion or evidence of MDR/RR-TB

H - isoniazid, R - Rifampicin, Z - pyrazinamide, E - Ethambutol, P - Rifapentine, M - Moxifloxacin

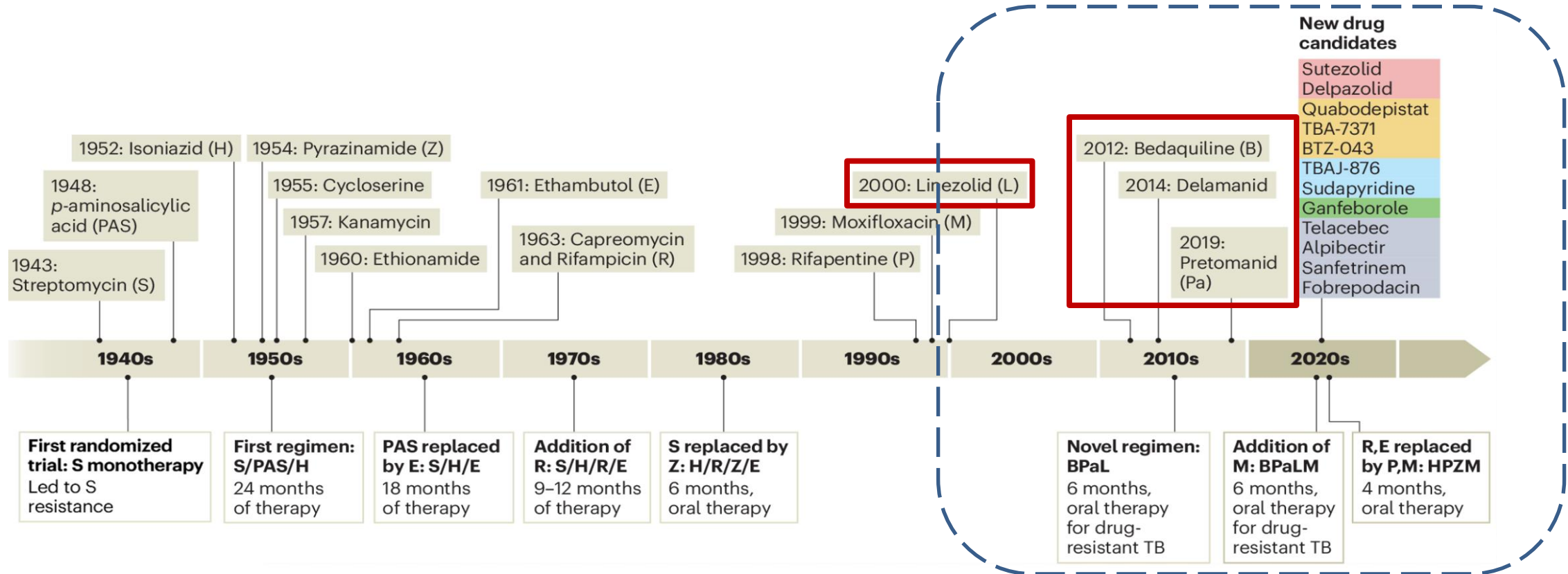
WHO consolidated guidelines on TB treatment, 2025

Drug resistant TB - definitions

TB disease caused by a strain of *Mycobacterium tuberculosis* complex that is resistant to any TB medicines.

- **Rifampicin-resistant TB (RR-TB):**
TB caused by *M. tb* strains that are resistant to rifampicin
- **Multidrug-resistant TB (MDR-TB):**
M. tb strains resistant to at least both isoniazid and rifampicin
- **Pre-extensively drug-resistant TB (pre-XDR-TB):**
MTBC with RR-TB (+/- isoniazid resistance), and resistant to at least one fluoroquinolone (either levofloxacin or moxifloxacin)
- **Extensively drug-resistant TB (XDR-TB):**
MTBC with RR-TB (+/- isoniazid resistance), and resistant to at least one fluoroquinolone (either levofloxacin or moxifloxacin) and to at least one other "Group A" drug (bedaquiline or linezolid)

Evolution of TB treatment drugs



Nuermberger, E.L., Chaisson, R.E. Restocking the tuberculosis drug arsenal. *Nat Med* (2024).

Treatment of MDR/RR TB using 6-month regimen

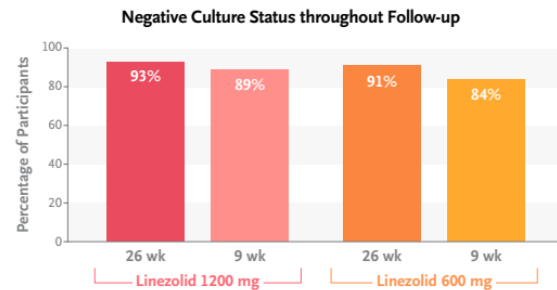
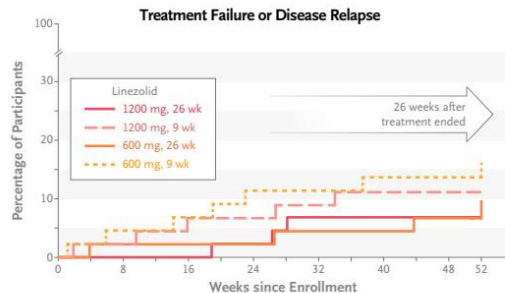
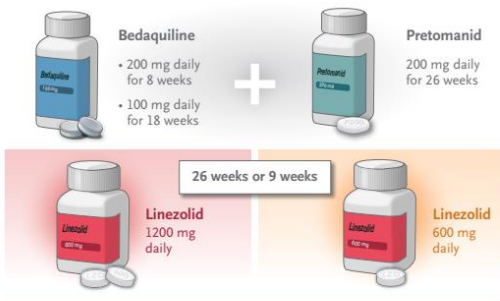
- Bedaquiline, pretomanid, linezolid (600 mg) and moxifloxacin **BPaLM**

The NEW ENGLAND JOURNAL of MEDICINE

RESEARCH SUMMARY

Bedaquiline–Pretomanid–Linezolid Regimens for Drug-Resistant Tuberculosis

Conradie F et al. DOI: 10.1056/NEJMoa2119430



CONCLUSIONS

In patients with highly drug-resistant tuberculosis, 600 mg of linezolid for 26 weeks resulted in a more favorable risk-benefit profile than other dose-duration regimens tested.

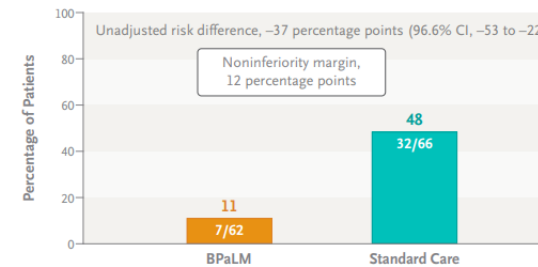
The NEW ENGLAND JOURNAL of MEDICINE

RESEARCH SUMMARY

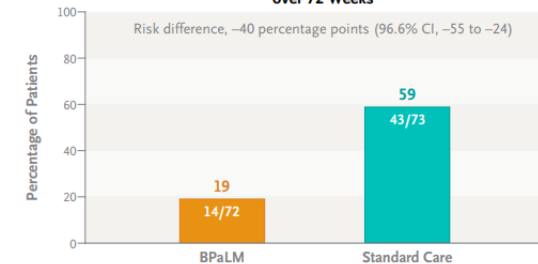
A 24-Week, All-Oral Regimen for Rifampin-Resistant Tuberculosis

Nyang'wa B-T et al. DOI: 10.1056/NEJMoa2117166

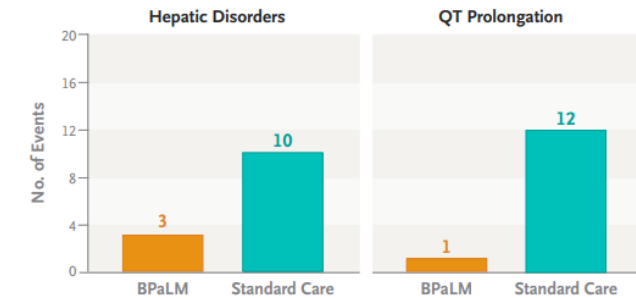
Unfavorable Outcome in Modified Intention-to-Treat Analysis



≥1 Serious Adverse Event or Adverse Event of Grade ≥3 over 72 Weeks



Most Common Adverse Events (Serious or Grade ≥3)



CONCLUSIONS

In patients with rifampin-resistant tuberculosis, a 24-week, all-oral BPaLM regimen was noninferior to the standard-care regimen with respect to unfavorable outcomes, and it had a better safety profile.

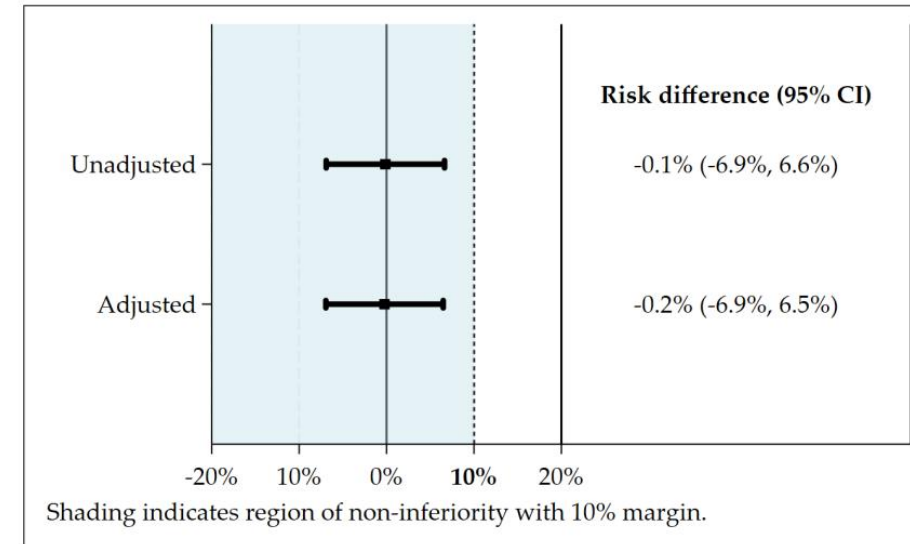
Treatment of MDR/RR TB using 6-month regimen

- Bedaquiline, delamanid, linezolid (600 mg), levofloxacin, and clofazimine in MDR/RR-TB patients with or without FQ resistance: **BDLLfxC**

BEAT Tuberculosis: a randomized controlled trial of a 6-month strategy for rifampicin-resistant tuberculosis

Francesca Conradie, Tasnim Badat, Asanda Poswa, Shakira Rajaram, Shaneen Kooverjee, Gary Maartens, Graeme Meintjes, Norbert Ndjeka, H.S Schaaf, Jennifer Hughes, Pauline Howell, Patrick Phillips

doi: <https://doi.org/10.1101/2025.05.04.25326549>



In place of 9-month or longer regimens in patients with MDR/RR-TB or pre-XDR-TB who are unable to receive pretomanid within the BPaL or BPaLM regimens.

Treatment of MDR/RR TB using modified 9-month regimen

EndTB trial multicountry: 9-month regimens

- **BLMZ**
- **BLLfxCZ**
- **BDLLfxZ**
- DIm-Cfz-Lzd-Lfx-Z
- DIm-Cfz-Mfx-Z

More options for treatment of RR-TB with three shortened, all oral regimens that were non-inferior to SoC when FQ resistance is excluded

Nine-month, all-oral regimens for rifampin-resistant tuberculosis

Guglielmetti L, Khan U, Velasquez GE, Gouillou M, Abubakirov A, Baudin E, Berikova E, Berry C, Bonnet M, Cellamare M, Chavan V, Cox V, Dakenova Z, de Jong BC, Ferlazzo G, Karabayev A, Kirakosyan O, Kiria N, Kunda M, Lachenal N, Lecca L, McIlleron H, Motta I, Mucching-Toscano S, Mushtaque H, Nahid P, Oyewusi L, Panda S, Patil S, Phillips P, Ruiz J, Salahuddin N, Sanchez-Garavito E, Seung KJ, Ticona E, Trippa L, Vargas D, Wasserman S, Rich ML, Varaine F, Mitnick

Evolution of DR TB treatment drugs

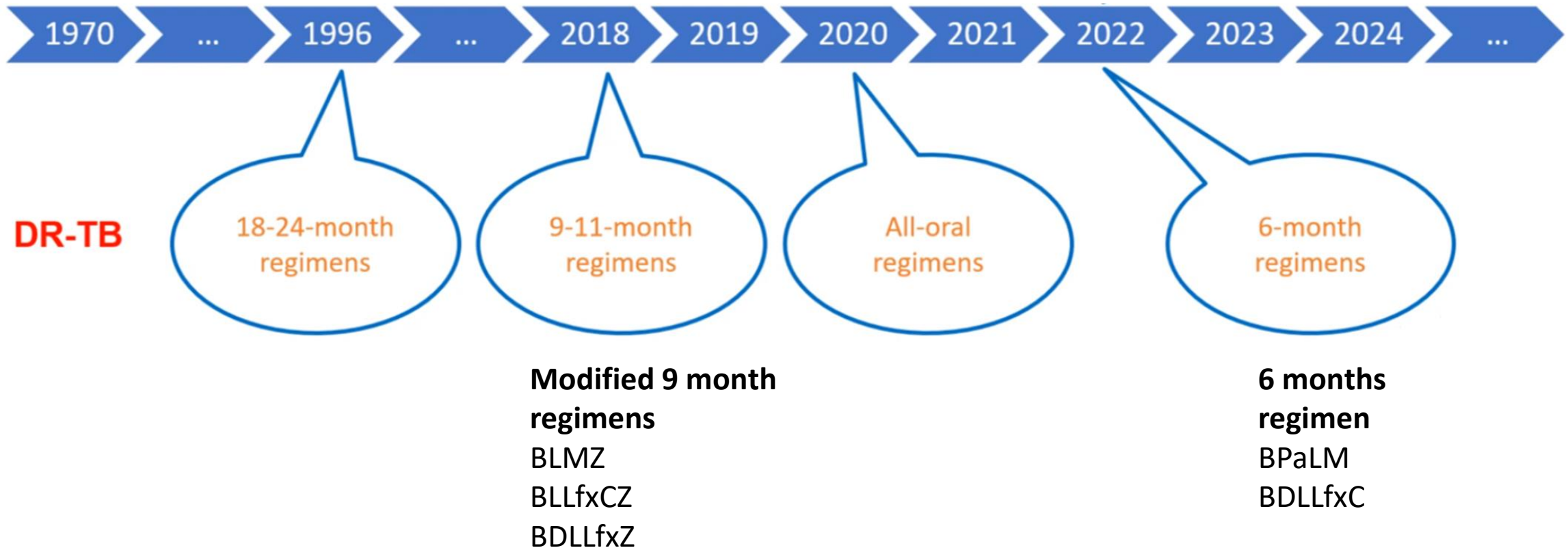
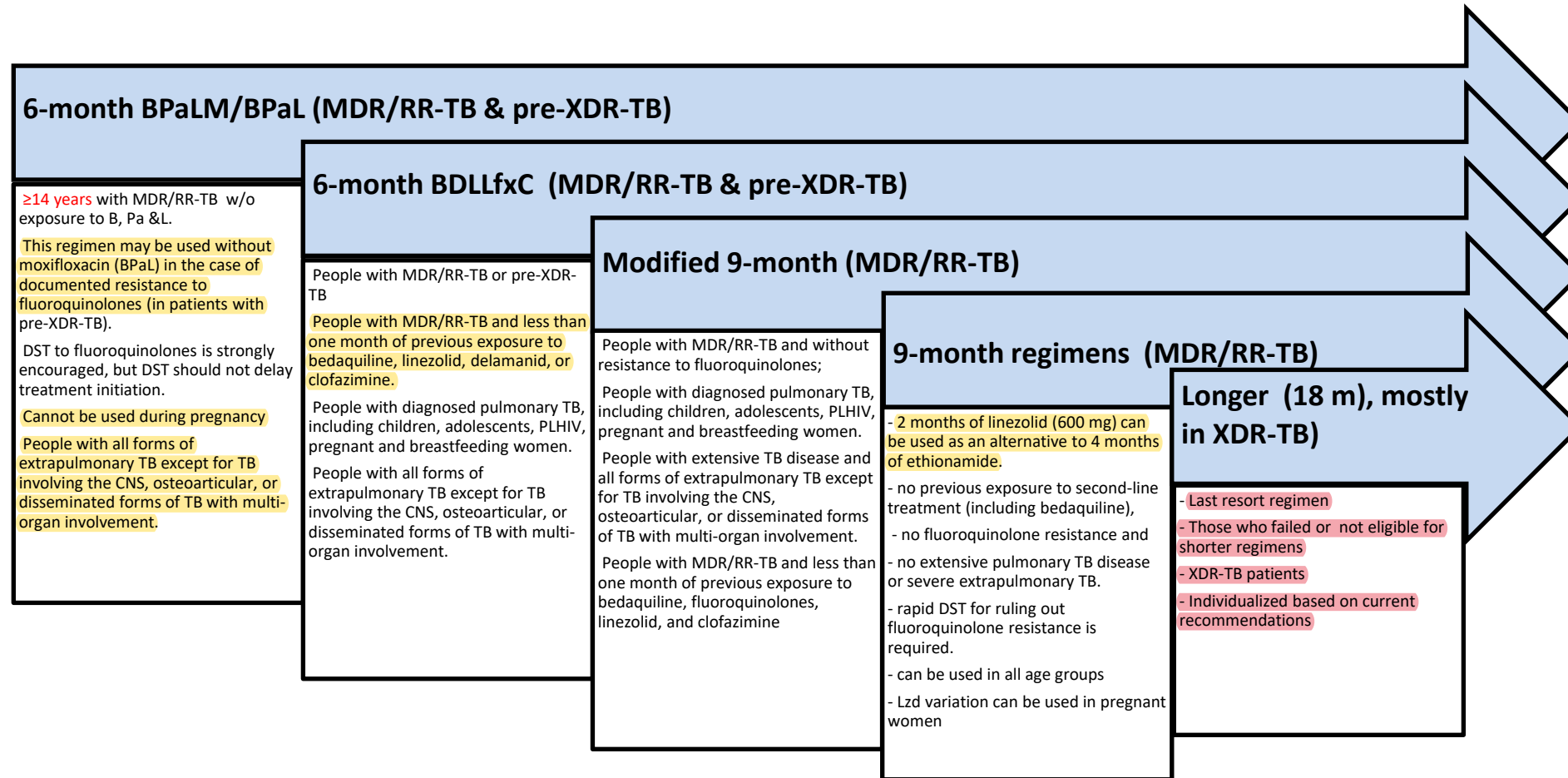


Fig: Adapted from overview of WHO consolidated guidelines on TB treatment, 2025

Treatment of DR TB



WHO guidelines & operational handbook: 2025
Module 4: Treatment and Care, Chapter 2: Drug-resistant TB treatment

Latent TB infection

- Tuberculosis infection (TBI) - state of persistent immune response to stimulation by *M. tb* antigens with no evidence of clinically manifest TB disease
- About one fourth of the world's population has been infected with TB
- TB preventive treatment (TPT) reduces the risk of TBI progressing to TB disease

WHO consolidated guidelines on TB: Module 1 TB preventive therapy, 2025

Target populations for TPT

WHO recommends assessing two main groups for TPT eligibility

- People at elevated risk of progression from infection to TB disease
- People at increased likelihood of exposure to TB disease

Populations eligible for TPT

People living with HIV

- Adults and adolescents
[regardless of ARV, pregnancy, previous TB treatment, immunosuppression and availability of test for TB infection]*
- Infants aged < 12 months who are in contact with TB*
- Children aged $\geq$ 12 months in a high TB transmission setting
- All children who successfully completed treatment for TB disease

WHO consolidated guidelines on TB: Module 1 TB preventive therapy, 2025

People living with HIV

- ~ 18 times more likely to develop TB disease than those without HIV infection (1)
- TB is the most frequent cause of AIDS-related deaths worldwide (2)
- TPT increases the survival of people with HIV even when they are on ART (3)
- TPT provides additional protection when given immediately after successful completion of TB treatment in people with HIV (4, 5)

1. Global tuberculosis report 2023; 2. Ford N et al. J Int AIDS Soc. 2016; 3. Badje A et al. Lancet Glob Health. 2017

4. Bruins WS et al. Infect Dis. 2017; 5. WHO Guidelines for intensified TB case-finding and IPT for PLHIV in resource-constrained settings, 2011

Populations eligible for TPT

Household contacts of pulmonary TB (bacteriologically confirmed)

- Children < 5 years*
- Individuals aged ≥ 5 years
- Exposed to MDR-TB

Other risk groups

- People who have silicosis, or who are initiating anti-TNF treatment or dialysis, or preparing for an organ or haematological transplant*
- Prisoners, health workers, immigrants from countries with a high TB burden, homeless people and PWUD

Household contacts (regardless of HIV status)

Children at highest risk

- < 5 years
 - higher risk of TBI and progressing rapidly to TB disease
 - With those < 2 years having a higher risk for severe/disseminated TB and mortality
- Risk of developing TB disease within 2 years: 18 - 19%
- Effectiveness of TPT in preventing TB disease
 - ~ 91% for children and adolescents with TBI

The risk of tuberculosis in children after close exposure: a systematic review and individual-participant meta-analysis



Leonardo Martinez, Olivia Cords, C Robert Horsburgh, Jason R Andrews, Pediatric TB Contact Studies Consortium*

Summary

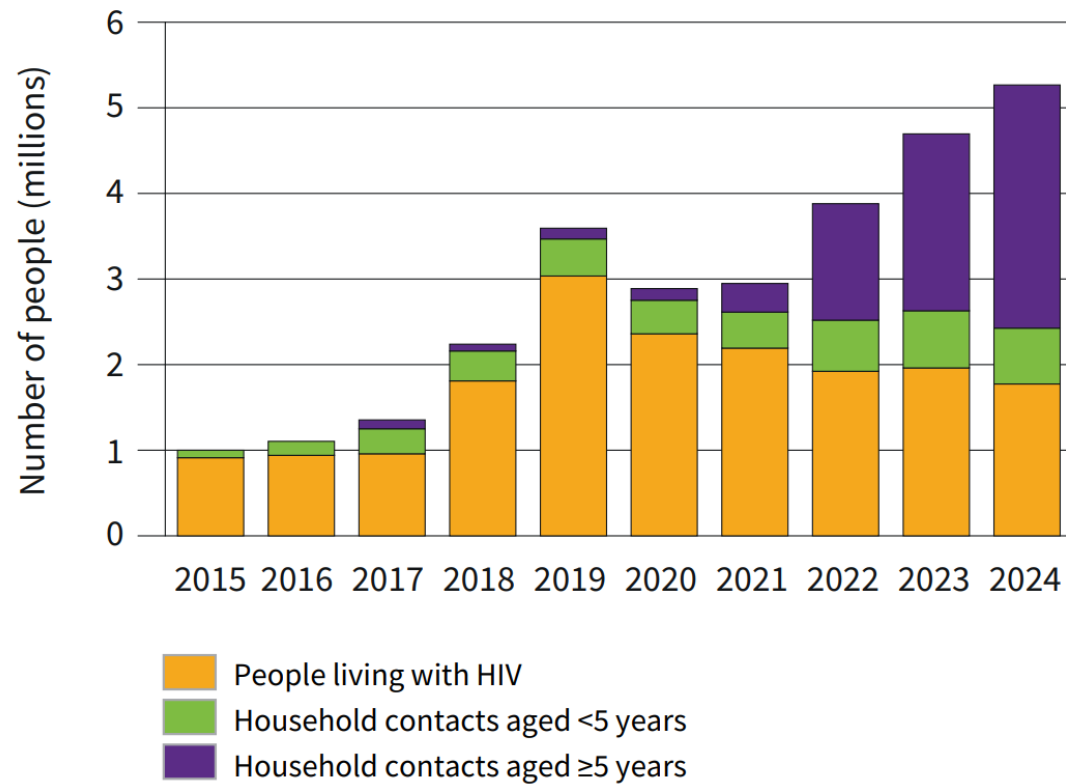
Background Tens of millions of children are exposed to *Mycobacterium tuberculosis* globally every year; however, there Lancet 2020; 395: 973-84

TPT is strongly recommended once TB disease has been ruled out

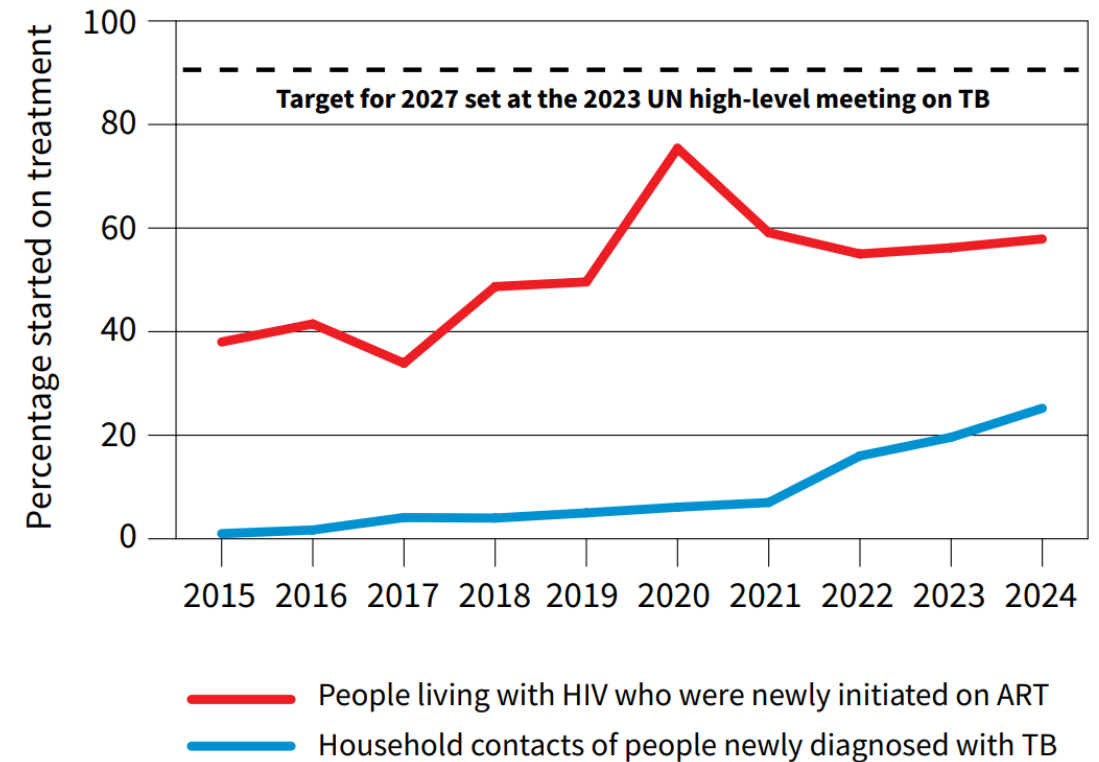
Martinez L. Lancet. 2020

Coverage of TB preventive therapy

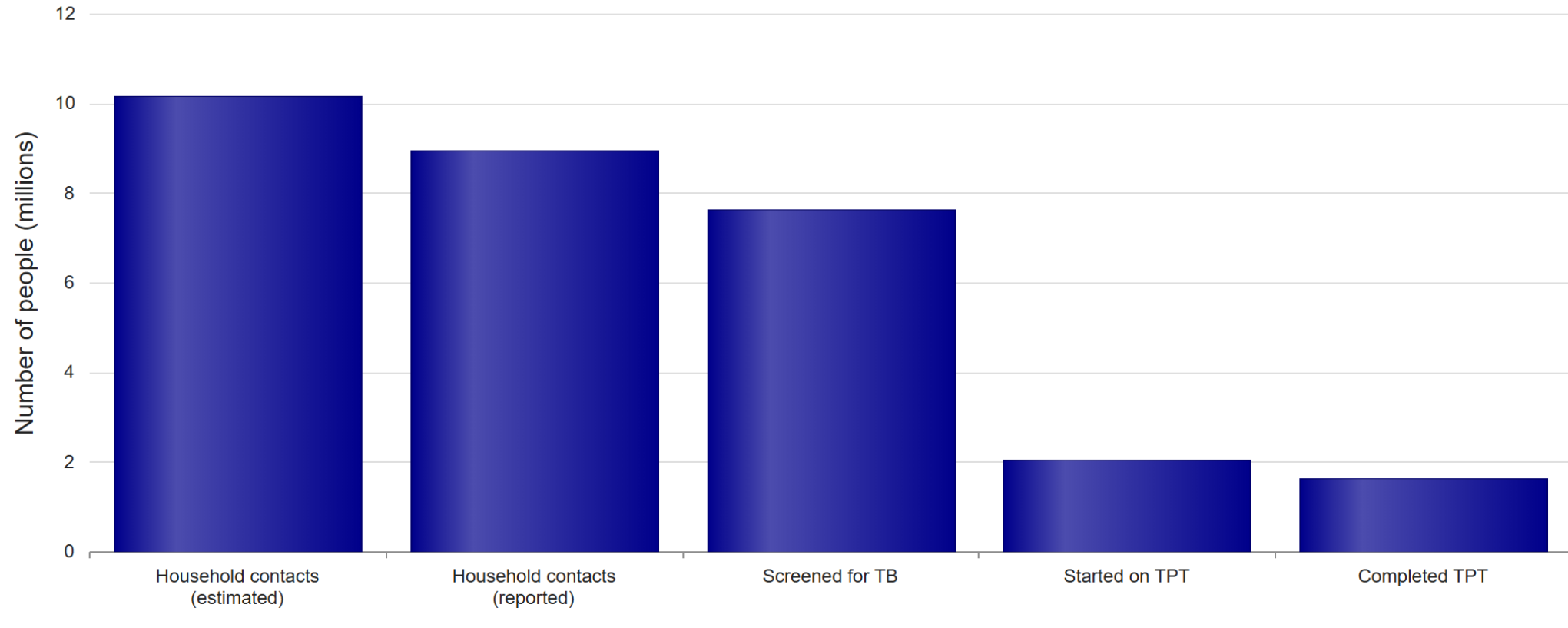
Global number of people provided with TPT, 2015–2024



Global coverage of TPT, 2015–2024



Cascade of care for provision of TPT among HHC of BcPTB, 2023



TPT options

- 1) 6 or 9-month, daily isoniazid*
- 2) 3-month, weekly isoniazid and rifapentine*
- 3) 3-month, daily isoniazid and rifampicin*
- 4) 1-month, daily isoniazid and rifapentine
- 5) 4-month, daily rifampicin
- 6) 6-month, daily levofloxacin after exposure to multidrug- or rifampicin-resistant TB*

WHO consolidated guidelines on TB: Module 1 TB preventive therapy, 2025

TB preventive therapy

3HP (PREVENT TB trial)

The NEW ENGLAND
JOURNAL *of* MEDICINE

ESTABLISHED IN 1812

DECEMBER 8, 2011

VOL. 365 NO. 23

Three Months of Rifapentine and Isoniazid for Latent Tuberculosis Infection

Timothy R. Sterling, M.D., M. Elsa Villarino, M.D., M.P.H., Andrey S. Borisov, M.D., M.P.H., Nong Shang, Ph.D., Fred Gordin, M.D., Erin Bliven-Sizemore, M.P.H., Judith Hackman, R.N., Carol Dukes Hamilton, M.D., Dick Menzies, M.D., Amy Kerrigan, R.N., M.S.N., Stephen E. Weis, D.O., Marc Weiner, M.D., Diane Wing, R.N., Marcus B. Conde, M.D., Lorna Bozeman, M.S., C. Robert Horsburgh, Jr., M.D., Richard E. Chaisson, M.D., for the TB Trials Consortium PREVENT TB Study Team*

1HP (BRIEF TB/A5279 trial)

The NEW ENGLAND
JOURNAL *of* MEDICINE

ESTABLISHED IN 1812

MARCH 14, 2019

VOL. 380 NO. 11

One Month of Rifapentine plus Isoniazid to Prevent HIV-Related Tuberculosis

S. Swindells, R. Ramchandani, A. Gupta, C.A. Benson, J. Leon-Cruz, N. Mwelase, M.A. Jean Juste, J.R. Lama, J. Valencia, A. Omoz-Oarhe, K. Supparatpinyo, G. Masheto, L. Mohapi, R.O. da Silva Escada, S. Mawlana, P. Banda, P. Severe, J. Hakim, C. Kanyama, D. Langat, L. Moran, J. Andersen, C.V. Fletcher, E. Nuermberger, and R.E. Chaisson, for the BRIEF TB/A5279 Study Team*

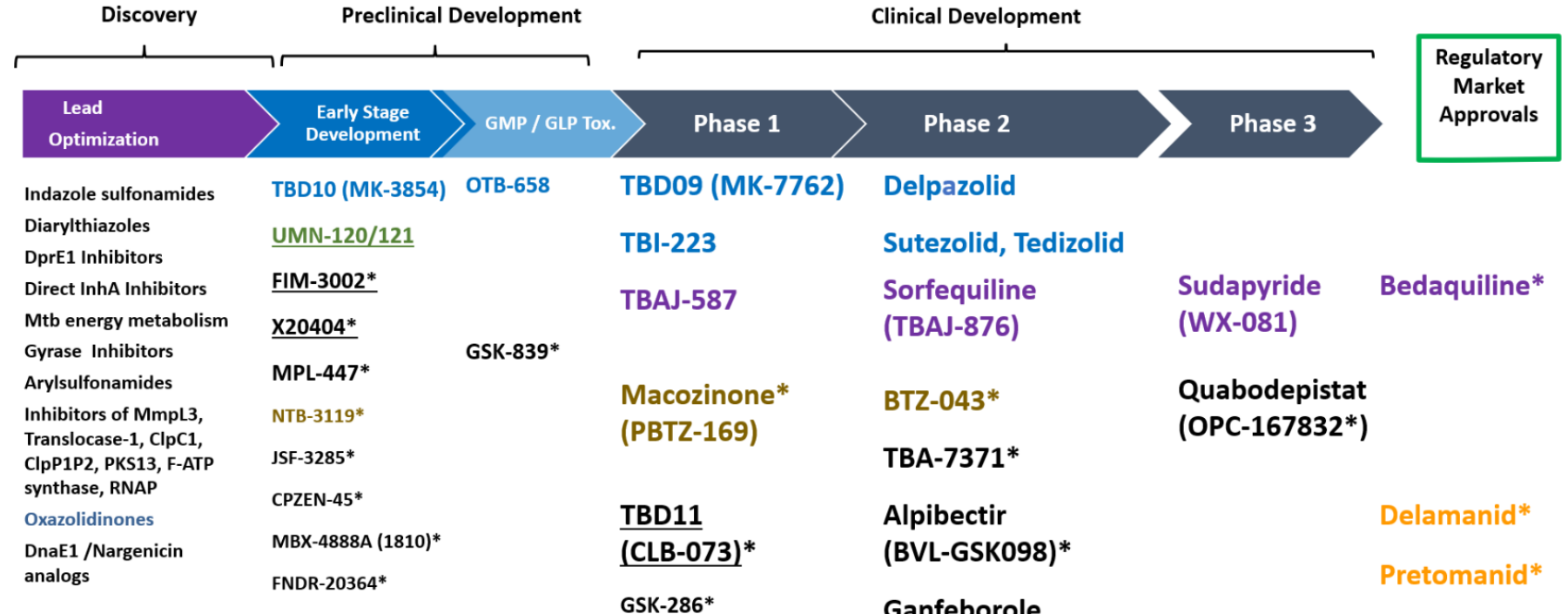
Advances in TB treatment

- Shift to shorter, all-oral regimens
- Improved safety and treatment completion
- Pan TB regimens
- Host directed therapies
- Long acting technologies
- Therapeutic vaccines

Advances in TB treatment

2025 Global New TB Drug Pipeline¹

- Shift to shorter, all-oral regimens
- Improved safety and treatment completion
- Pan TB regimens



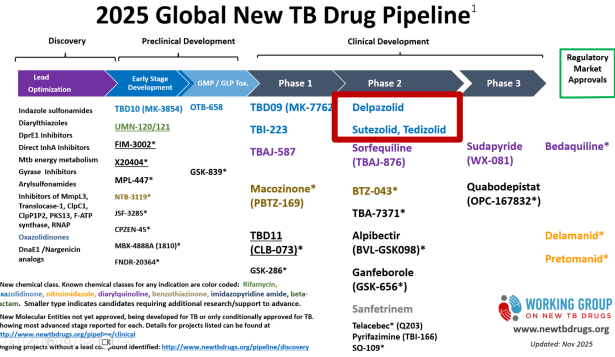
*New chemical class. Known chemical classes for any indication are color coded: Rifamycin, oxazolidinone, nitroimidazole, diarylquinoline, benzothiazinone, imidazopyridine amide, beta-lactam. Smaller type indicates candidates requiring additional research/support to advance.

¹ New Molecular Entities not yet approved, being developed for TB or only conditionally approved for TB. Showing most advanced stage reported for each. Details for projects listed can be found at <http://www.newtbdrugs.org/pipeline/clinical>

Ongoing projects without a lead compound identified: <http://www.newtbdrugs.org/pipeline/discovery>

Advances in TB treatment

Oxazolidinones



Delpazolid in combination with bedaquiline, delamanid, and moxifloxacin for pulmonary tuberculosis (PanACEA-DECODE-01): a prospective, randomised, open-label, phase 2b, dose-finding trial



Lilian Tina Minja, Isabella van der Feltz, Christina Manyama, Stellah Mpagama, Francis Mhimira, Ivan Noreña, Modulakgotla Sebe, Mohammed Rassool, Robert S Wallis, Nyanda Ntinginya, Alphonse Lijoyo, Beno Mbeya, Larissa Wagnerberger, Tresphory Zumba, Daud D Peter, Heeran Makkan, Derek J Sloan, Lindsey te Brake, Jodie A Schildkraut, Rob Aarnoutse, Timothy D McHugh, Leticia Wildner, Martin J Boeree, Lawrence Geiter, Young-Lag Cho, Brian H Aldana, Patrick P J Phillips, Michael Hoelscher, Elin M Svensson, Norbert Heinrich, on behalf of the PanACEA consortium*



Summary
Background Linezolid plays a crucial role in the first-line treatment of drug-resistant tuberculosis globally. Its prolonged use can lead to neurological and haematological toxicity, highlighting the need for safer oxazolidinones. Delpazolid, a novel oxazolidinone, might be safer. We aimed to evaluate the safety and efficacy of delpazolid and identify an optimal dose.

Lancet Infect Dis 2025; 25: 1219–29
 Published Online July 8, 2025
[https://doi.org/10.1016/S1473-3099\(25\)00289-0](https://doi.org/10.1016/S1473-3099(25)00289-0)



Sutezolid in combination with bedaquiline, delamanid, and moxifloxacin for pulmonary tuberculosis (PanACEA-SUDOCU-01): a prospective, open-label, randomised, phase 2b dose-finding trial



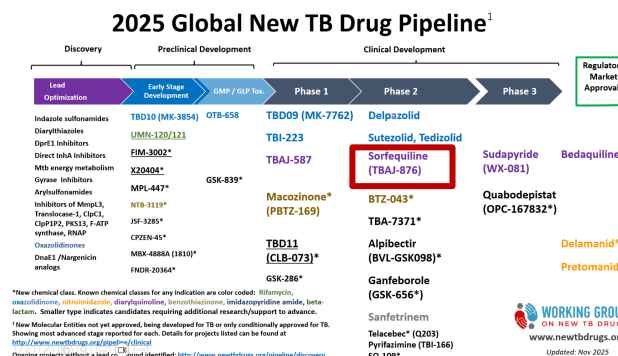
Norbert Heinrich*, Christina Manyama*, Simon E Koele, Stellah Mpagama, Francis Mhimira, Modulakgotla Sebe, Robert S Wallis, Nyanda Ntinginya, Alphonse Lijoyo, Beno Huglin, Lilian Tina Minja, Larissa Wagnerberger, Krista Stoycheva, Tresphory Zumba, Ivan Noreña, Daud D Peter, Heeran Makkan, Derek J Sloan, Lindsey te Brake, Jodie Schildkraut, Rob E Aarnoutse, Timothy D McHugh, Leticia Wildner, Martin Boeree, Brian H Aldana, Patrick P J Phillips, Michael Hoelscher, Elin M Svensson, on behalf of the PanACEA consortium†

Summary
Background Linezolid is a key component globally in first-line therapy for drug-resistant tuberculosis but has considerable toxicity. New and safer alternative oxazolidinones are needed. Sutezolid is one such promising alternative. We aimed to evaluate preliminary efficacy and safety of sutezolid and to identify an optimal dose.

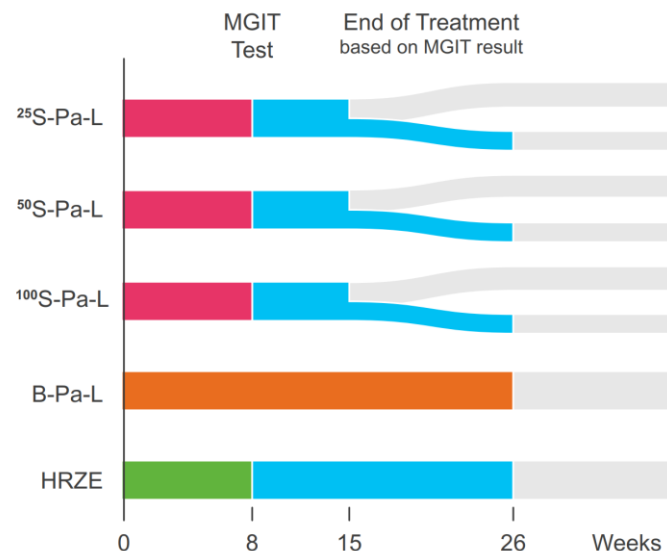
Lancet Infect Dis 2025; 25: 1208–18
 Published Online July 8, 2025

Advances in TB treatment

Diarylquinolones



The NC-009 trial (a pan-Phase 2 clinical trial) showed that, overall, sorfequiline had greater activity than bedaquiline. The 100 mg SPaL regimen had greater activity against TB than the standard of care HRZE (isoniazid, rifampin, pyrazinamide, and ethambutol), indicating the potential to shorten treatment time for active TB. In addition, the SPaL regimen had a comparable safety profile to the standard of care for people with drug-sensitive TB (DS-TB).



SPaL BPaL HRZE HR

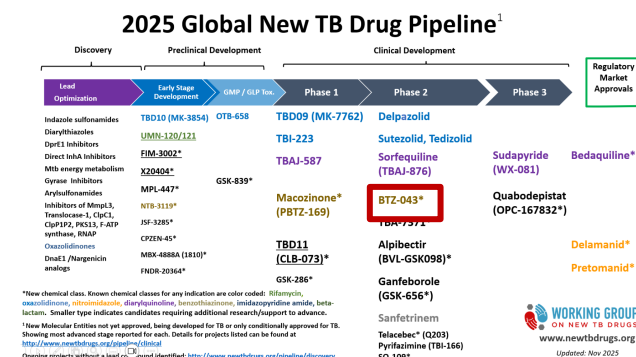
COPENHAGEN (November 19, 2025)



<https://doi.org/10.1186/s13063-025-09413-5>

Advances in TB treatment

DprE1 inhibitors

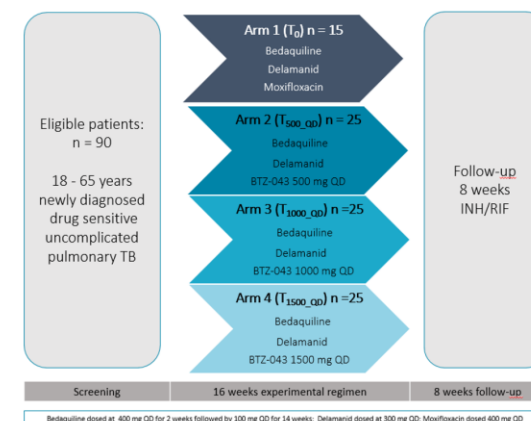


BTZ-043 Dose Evaluation in Combination and Selection (DECISION)

ClinicalTrials.gov ID  NCT05926466

Sponsor  Michael Hoelscher

Information provided by  Michael Hoelscher, Ludwig-Maximilians - University of Munich (Responsible Party)



Safety and Efficacy Evaluation of 4-month Regimen of OPC-167832, Delamanid and Bedaquiline in Participants With Drug-Susceptible Pulmonary TB

ClinicalTrials.gov ID  NCT05221502

Sponsor  Otsuka Pharmaceutical Development & Commercialization, Inc.

Information provided by  Otsuka Pharmaceutical Development & Commercialization, Inc. (Responsible Party)

Advances in TB treatment

Recruiting 

Platform Assessing Regimens and Durations In a Global Multisite Consortium for TB (PARADIGM4TB)

ClinicalTrials.gov ID  NCT06114628

Sponsor  University College, London

Information provided by  University College, London (Responsible Party)

Last Update Posted  2025-10-06

Ganfeborole (GSK)
BTZ-043 (LMU)

Quabodepistat (Otsuka)
Delpazolid (LigaChem
Biosciences Inc)

Licensed drugs:
Bedaquiline, linezolid,
moxifloxacin, pyrazinamide,
delamanid/pretomanid,

Advances in TB prevention



United States of America



BREACH-TB Trial

***Bedaquiline Roll-out Evidence in Contacts
and People Living with HIV to Prevent TB***

BREACH-TB is a Phase 2/3, open-label, multicenter,
randomized controlled, non-inferiority trial



Advances in TB prevention and treatment

- Long acting therapeutics
 - Low water solubility
 - Low clearance in plasma
 - High drug potency

- Examples

- LA Bedaquiline
- LA rifapentine/isoniazid
- Long acting rifamycins (rifapentine, rifabutin)
- Combination long acting regimens

Long-Acting Technologies for TB

TB Alliance presented early preclinical data on the efficacy of LA formulations of pretomanid and TBAJ-876 (sorfequiline) for TB treatment.

Begum N et al. Poster Presented at CROI 202625, Feb 2026

Published in final edited form as:

Int J Tuberc Lung Dis. 2018 February 01; 22(2): 125–132. doi:10.5588/ijtld.17.0486.

LONG-ACTING FORMULATIONS FOR THE TREATMENT OF LATENT TUBERCULOSIS INFECTION: OPPORTUNITIES AND CHALLENGES

Susan Swindells¹, Marco Siccardi², Stephanie E. Barrett³, David B. Olsen⁴, Jay A. Grobler⁴, Anthony Podany¹, Eric Nuermberger⁵, Peter Kim⁶, Clif Barry⁷, Andrew Owen², Daria Hazuda⁴, and Charles Flexner⁵

Host directed therapies

Tian et al.
European Journal of Medical Research (2025) 30:267
<https://doi.org/10.1186/s40001-025-02443-4>

European Journal
of Medical Research



REVIEW

Open Access

Host-directed therapy for tuberculosis

Na Tian^{1†}, Hongqian Chu^{2†}, Qi Li¹, Hong Sun², Jingfang Zhang², Naihui Chu^{1*} and Zhaogang Sun^{2*}



- Target **host immune pathways rather than the bacterium**
- Aim to enhance immune control of infection and improve treatment outcomes in TB

Future TB treatment may combine antimicrobial therapy with host-directed therapies to optimize cure and reduce lung damage

<https://doi.org/10.1186/s40001-025-02443-4>

Advances in TB vaccines

The NEW ENGLAND JOURNAL of MEDICINE

ORIGINAL ARTICLE

Phase 2b Controlled Trial of M72/AS01_E Vaccine to Prevent Tuberculosis

O. Van Der Meeren, M. Hatherill, V. Nduba, R.J. Wilkinson, M. Muyoyeta,
E. Van Brakel, H.M. Ayles, G. Henostroza, F. Thienemann, T.J. Scriba, A. Diacon,
G.L. Blatner, M.-A. Demoitié, M. Tameris, M. Malahleha, J.C. Innes, E. Hellström,
N. Martinson, T. Singh, E.J. Akite, A. Khatoon Azam, A. Bollaerts, A.M. Ginsberg,
T.G. Evans, P. Gillard, and D.R. Tait

- Phase 2b, randomized, double-blind, placebo-controlled trial
- 18 to 50 years (HIV negative)
- TBI (IGRA)
- Two doses of either M72/AS01_E or placebo intramuscularly 1 month apart
- Follow up - 2.3 years

Conclusions: M72/AS01_E provided 54.0% protection for *M. tb*-infected adults against active PTB, without evident safety concerns

Advances in TB vaccines

Live-attenuated *Mycobacterium tuberculosis* vaccine, MTBVAC, in adults with or without *M tuberculosis* sensitisation: a single-centre, phase 1b–2a, double-blind, dose-escalation, randomised controlled trial

Angelique Kany Kany Luabeya*, Virginie Rozot*, Claire Imbratta*, Frances Ratangee, Justin Shenje, Michele Tameris, Simon C Mendelsohn, Hennie Geldenhuys, Michelle Fisher, Munyaradzi Musvosvi, Carly Young, Humphrey Mulenga, Nicole Bilek, Simbarashe Mabwe, Ingrid Murillo Jelsbak, Esteban Rodríguez, Eugenia Puentes, Juana Doce, Nacho Aguilo, Carlos Martin, Cadwill Pillay, Dereck Tait, Marisa Russell, Arrie Van Der Merve, Kathryn Rutkowski, Devin Hunt, Ann Ginsberg, Thomas J Scriba, Mark Hatherill, the A050 team

Summary

Background An effective adult vaccine is needed to control tuberculosis. We evaluated the safety and immunogenicity of a live-attenuated *Mycobacterium tuberculosis* vaccine (MTBVAC).



Lancet Glob Health 2025;
13: e1030–42

- Phase 1b–2a, double-blind, randomized, dose-escalation trial
- Evaluating safety and immunogenicity of the live-attenuated TB vaccine MTBVAC compared with BCG revaccination
- 18–50 years (HIV-negative)
- Previously received BCG

MTBVAC showed similar safety and reactogenicity and greater immunogenicity when compared to BCG

Advances in TB treatment trials

- MAMS designs
- Regimen-based trials
- Dose optimization
- Seamless Phase II/III
- Platform trials

Collaborative research consortia

Thank you